

TINGXI LI

800 W Campbell Road, Richardson, TX 75080

☎ (213)-795-5275 ✉ tingxi.li@utdallas.edu [in](#) [linkedin](#) [🏠 homepage](#)

Education

University of Texas at Dallas

Doctor of Philosophy in Computer Science

Sep. 2024 – Present

Dallas, TX

Dalian University of Technology

Bachelor of Science in Chemistry

Sep. 2019 – May 2024

Dalian, China

Technical University of Munich

Visiting Student: Computer Science

Apr. 2022 – Oct. 2022

Munich, Germany

Academic Experience

Adversarial Attack on a Robotic Arm

Jan. 2024 – Present

UT Dallas | Supervisor: Wei Yang

Dallas, TX

- Replaced the non-differentiable heightmap renderer with a differentiable one, enabling gradient computation through the entire model. This enhancement facilitates direct optimization of the model using backpropagation algorithms.
- For the object grasping task investigation, experiments conducted within the Bulletarm robotic framework demonstrated that the effect of a given action on the environment is deterministic. Moreover, it was observed that the boundaries between success and failure within the action space are non-robust.

Survey of Efficiency Robustness of Dynamic Deep Learning Systems

Jun. 2024 – Present

UT Dallas | Supervisor: Wei Yang

Dallas, TX

- Wrote a section of the survey paper, introducing and categorizing existing efficiency attacks on dynamic deep learning systems, providing a structured analysis to enhance understanding of potential vulnerabilities.

Efficiency Attack on Multi-level Applications

Jun. 2024 – Present

UT Dallas | Supervisor: Wei Yang

Dallas, TX

- Applied efficiency attacks on multi-level applications, evaluating their impact on downstream tasks by measuring energy consumption and time delays.
- Proposed and developed defense mechanisms to detect and mitigate these attacks, enhancing system resilience.

Industrial Experience

Sophgo

June. 2024 – Aug. 2024

Intern

Shenzhen, China

- Developed and implemented API code in C++ for models deployed on RISC-V processors. Authored comprehensive documentation to support efficient deployment and usage.
- Fine-tuned models on private datasets, identified failure cases, and implemented data augmentation strategies.

Projects

Amazon Trusted AI Competition

Nov. 2024 – Present

Team Member

Seattle, WA

- Finalist team in the Amazon Trusted AI Competition, focused on jailbreaking CodeLLMs under a black-box setting.
- Fine-tuned surrogate models to evaluate and guide adversarial prompt generation, achieving 0.80+ recall and 0.75+ F1-score in malicious intent detection.
- Built an iterative model refinement pipeline to continuously integrate new attack data and improve surrogate model accuracy.

Publications

SoK: Efficiency Robustness of Dynamic Deep Learning Systems

USENIX Security 2025

Ravishka Rathnasuriya, Tingxi Li, Zexin Xu, Zihe Song, Mirazul Haque, Simin Chen, Wei Yang

- Co-authored a systematization-of-knowledge (SoK) paper surveying efficiency attacks on dynamic DNN systems. Categorized attack strategies, reproduced published results, and benchmarked defenses in unstudied scenarios.

Miscellaneous

Research Interest: AI Security; Software Engineering; Deep Learning Compilers

Tech Stack: Python; C; C++; Java; PyTorch; LaTeX; SQL